MATH423: Introduction to Computational Topology
Lecture Notes

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2026-27 FALL

1 Abstract and Geometric Simplicial Complexes

In the first week, we briefly introduced geometric simplices. This week, we will formalize abstract simplicial complexes, which strip away the geometry to leave only combinatorial data. This combinatorial approach is essential for representing topological spaces algorithmically. We will then introduce the standard methods for building complexes from point cloud data: the Čech and Vietoris-Rips complexes.

It helps to keep three levels of description apart:

$$\underbrace{X}_{\text{a topological space}} \xleftarrow{\cong} \underbrace{|K|}_{\text{a polyhedron in } \mathbb{R}^n} \longleftarrow \underbrace{K}_{\text{a finite list of vertex sets}}$$

The space X is what we want to understand, for example a surface or the unknown shape from which a data set was sampled. The polyhedron $|K|$ is a concrete model of X built from geometric simplices. The abstract complex K is the data structure that a computer actually stores. Last week we moved from the first level to the second; this week we move from the second to the third.

1.1 Abstract Simplicial Complexes

While a geometric simplicial complex lives in $\mathbb{R}^n$ and consists of actual geometric objects (points, line segments, solid triangles), an abstract simplicial complex is a purely combinatorial object.

Why forget the coordinates?

A geometric simplicial complex carries more information than topology needs. Its topology depends only on which sets of vertices span simplices, not on where the vertices are placed. For instance, a hollow triangle drawn with an equilateral or with a very thin shape is homeomorphic to S^1 either way. In general, if the vertices of two geometric complexes can be matched so that a set of vertices spans a simplex in one complex exactly when the matching set spans a simplex in the other, then extending the matching linearly over each simplex gives a homeomorphism between their polyhedra. Dropping the coordinates has several practical advantages.

- **Exactness.** Condition 2 in the definition of a geometric simplicial complex requires computing intersections of simplices in $\mathbb{R}^n$. With coordinates stored as floating point numbers, deciding whether two triangles meet only along an edge or overlap slightly can go wrong. The conditions in the abstract definition below are statements about finite sets of integers, and they are checked exactly.

- **No ambient space needed.** Many complexes are not naturally drawn anywhere. The real projective plane cannot be embedded in $\mathbb{R}^3$ at all, so a geometric triangulation of it must live in $\mathbb{R}^4$ or higher. The co-authorship complex of a department, or a complex built from a table of distances between DNA sequences, has no natural coordinates at all.

- **Size.** A list of vertex sets is the smallest description from which the topology can be recovered. Coordinates can be added later, if a picture is needed.

The passage from geometry to combinatorics is made by recording only the vertex sets. If K is a geometric simplicial complex with vertex set V, its **vertex scheme** is the collection of all subsets of V that span a simplex of K. For example, the vertex scheme of the boundary of the tetrahedron consists of the four triples listed in the previous week, together with their 6 subsets of size two and 4 subsets of size one. Labelling the vertices of the hollow triangle K' of Week 1 by $1, 2, 3$, its vertex scheme consists of $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$, and the vertex scheme of the solid triangle additionally contains $\{1, 2, 3\}$.

Any vertex scheme has two evident properties. Every vertex is a 0-simplex, so every singleton $\{v\}$ belongs to it. Every face of a simplex is again a simplex of K, so every non-empty subset of a set in the vertex scheme belongs to it as well. The second condition in the definition of a geometric complex, about intersections, has no counterpart: it is automatically encoded by the fact that simplices are recorded through their vertex sets. The definition of an abstract simplicial complex simply takes these two properties as axioms.

Definition 1.1. An **abstract simplicial complex** is a pair (V, K) where V is a set of vertices and K is a family of non-empty finite subsets of V such that:

1. For every $v \in V$, the singleton $\{v\}$ is in K.

2. If $\sigma \in K$ and $\tau \subseteq \sigma$ is non-empty, then $\tau \in K$.

The sets $\sigma \in K$ are called **simplices** or faces. The **dimension** of a simplex σ is defined as $|\sigma| - 1$. The dimension of the complex K is the maximum dimension of any of its simplices.

A simplex that is not a proper face of any other simplex is called **maximal** (or a facet of K). Since K is closed under taking subsets, a finite complex is completely determined by its list of maximal simplices; this is how complexes are usually stored in software.

Example 1.2.

- The **full k-simplex** Δ^k on $V = \{0, 1, \ldots, k\}$ consists of all non-empty subsets of V. It has $2^{k+1} - 1$ simplices.

- The **boundary of the tetrahedron** on $V = \{1, 2, 3, 4\}$ has maximal simplices $\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}$. It is Δ^3 with the single 3-simplex $\{1, 2, 3, 4\}$ removed, and it triangulates S^2.

- The family $\{\{1\},\{2\},\{3\},\{1,2\},\{1,2,3\}\}$ is *not* a simplicial complex: the subsets $\{1,3\}$ and $\{2,3\}$ of $\{1,2,3\}$ are missing.
- Any graph without loops or multiple edges is a 1-dimensional abstract simplicial complex.

Every geometric simplicial complex naturally defines an abstract simplicial complex by simply listing the vertices that span each simplex. Conversely, every abstract simplicial complex of dimension d has a **geometric realization** in $\mathbb{R}^{2d+1}$.

Example 1.3. The bound $2d+1$ cannot be improved in general. For $d=1$, the complete graph K_5 and the complete bipartite graph $K_{3,3}$ are 1-dimensional complexes that cannot be realized in $\mathbb{R}^2$ (Kuratowski), but every graph can be realized in $\mathbb{R}^3$: place the vertices at distinct points of the moment curve $t \mapsto (t, t^2, t^3)$. No four points on this curve are coplanar, so two edges with four distinct endpoints never meet.

Several standard subcomplexes are attached to a simplex or a vertex.

Definition 1.4. Let K be an abstract simplicial complex and $\sigma \in K$.

- The **k-skeleton** $K^{(k)}$ is the subcomplex of all simplices of dimension at most k.
- The **star** of σ is $\mathrm{st}(\sigma) = \{\tau \in K \mid \sigma \subseteq \tau\}$, the set of simplices having σ as a face. In general it is not a subcomplex; see Proposition 1.6.
- The **closed star** $\overline{\mathrm{st}}(\sigma)$ is the smallest subcomplex containing $\mathrm{st}(\sigma)$, i.e., all faces of simplices in the star.
- The **link** of σ is $\mathrm{lk}(\sigma) = \{\tau \in \overline{\mathrm{st}}(\sigma) \mid \tau \cap \sigma = \emptyset\}$. For a vertex v this is $\mathrm{lk}(v) = \{\tau \in K \mid v \notin \tau,\ \tau \cup \{v\} \in K\}$.

Recall that a **subcomplex** of K is a subset $L \subseteq K$ which is itself a simplicial complex, that is, which contains every non-empty face of each of its simplices. The three sets in the definition behave differently with respect to this condition. The star is built by going *up* from σ: it collects the simplices that contain σ. Being a subcomplex requires closure going *down*, under taking faces. The faces of a simplex $\tau \in \mathrm{st}(\sigma)$ that do not contain σ are missing from the star, and this is the only reason it fails to be a subcomplex. The closed star puts these faces back, and the link consists of those faces that are disjoint from σ.

Example 1.5. Let K be the boundary of the tetrahedron on $\{1,2,3,4\}$ and $v=1$.

- **Star.** The simplices containing 1 are

$$\mathrm{st}(1) = \big\{\{1\},\ \{1,2\},\{1,3\},\{1,4\},\ \{1,2,3\},\{1,2,4\},\{1,3,4\}\big\},$$

  seven simplices in total. This is not a subcomplex: the triangle $\{1,2,3\}$ belongs to $\mathrm{st}(1)$, but its edge $\{2,3\}$ does not, because $1 \notin \{2,3\}$. For the same reason the vertices $\{2\},\{3\},\{4\}$ are missing, although they are faces of the edges $\{1,2\},\{1,3\},\{1,4\}$. Thus $\mathrm{st}(1)$ contains triangles whose boundary edges are not all present, which a simplicial complex never does.

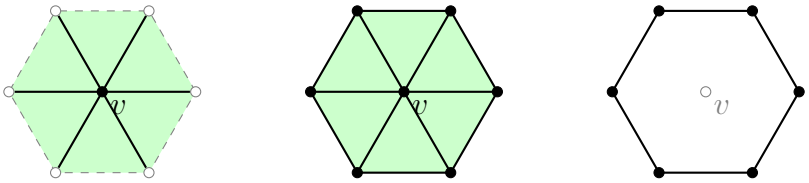
- **Closed star.** Adding all faces of the simplices in st(1) adds exactly the missing ones,

$$\{2\},\{3\},\{4\},\ \{2,3\},\{2,4\},\{3,4\},$$

giving a subcomplex with 13 simplices. Its polyhedron consists of the three triangles around the vertex 1, a triangulated disk.

- **Link.** The simplices of the closed star disjoint from $\{1\}$ are exactly the six simplices just added, so lk(1) is the hollow triangle on $\{2,3,4\}$, a circle. It is the boundary of the disk $|\overline{\mathrm{st}}(1)|$.

For a vertex the closed star is the disjoint union $\overline{\mathrm{st}}(v) = \mathrm{st}(v) \sqcup \mathrm{lk}(v)$: a face of a simplex containing v either contains v, and then lies in the star, or does not, and then lies in the link. That the link of a vertex in a closed surface is a circle is a general fact; see Worksheet 1, Problem 6.



(a) $\mathrm{st}(v)$ (b) $\overline{\mathrm{st}}(v)$ (c) $\mathrm{lk}(v)$

Figure 1: A vertex v of a triangulated surface lying in six triangles. (a) The star: the triangles and edges containing v, together with v; the outer edges and outer vertices (dashed and hollow) are missing, so the star is not closed under taking faces. (b) The closed star: the missing faces are added, giving a disk. (c) The link: the simplices of the closed star not containing v, a hexagon.

Proposition 1.6. *Let $\sigma \in K$. Then $\mathrm{st}(\sigma)$ is a subcomplex of K if and only if $\sigma = \{v\}$ is a vertex which is not a face of any other simplex of K, in which case $\mathrm{st}(\sigma) = \{\{v\}\}$.*

Proof. If $\sigma = \{v\}$ lies in no other simplex, then $\mathrm{st}(\sigma) = \{\{v\}\}$, which is a subcomplex. Conversely, suppose $\mathrm{st}(\sigma)$ is a subcomplex. Since $\sigma \in \mathrm{st}(\sigma)$, every non-empty face of σ lies in $\mathrm{st}(\sigma)$ and hence contains σ; so σ has no proper non-empty face, that is, $\sigma = \{v\}$ is a vertex. If v were a face of some simplex $\tau \neq \{v\}$, then $\tau \setminus \{v\}$ would be a non-empty face of $\tau \in \mathrm{st}(v)$ not containing v, so it would not lie in $\mathrm{st}(v)$, a contradiction. □

In other words, apart from isolated vertices, the star is *never* a subcomplex. The reason to consider it anyway is geometric. The union of the interiors of the simplices in $\mathrm{st}(\sigma)$, called the **open star**, is an open subset of $|K|$ containing the interior of σ. It is the natural neighborhood of σ in $|K|$, as in Figure (a) above, and open sets are not unions of closed simplices. The closed star is its closure, and for a vertex v the link is what is added in passing from the open star to its closure, namely its frontier.

Example 1.7. The link is defined for simplices of every dimension. Let K again be the boundary of the tetrahedron and $\sigma = \{1,2\}$. Then $\mathrm{st}(\sigma) = \big\{\{1,2\},\{1,2,3\},\{1,2,4\}\big\}$, and the closed star consists of the two triangles $\{1,2,3\},\{1,2,4\}$ together with all their faces, 9 simplices in all. Among these, the ones disjoint from $\{1,2\}$ are $\{3\}$ and $\{4\}$;

the edge $\{3,4\}$ is not in the closed star. Hence $\mathrm{lk}(\{1,2\}) = \big\{\{3\},\{4\}\big\}$, two points, which correspond to the two triangles on either side of the edge. Notice that for an edge the closed star is *not* the disjoint union of the star and the link: the simplices $\{1\},\{2\},\{1,3\},\{2,3\},\{1,4\},\{2,4\}$ lie in the closed star but meet σ without containing it, so they belong to neither. The link of the edge appears in the link condition for edge contractions in Section 4.

Definition 1.8. Let K be a finite abstract simplicial complex with f_k simplices of dimension k. The vector $(f_0, f_1, \ldots, f_d)$ is the **f-vector** of K, and its **Euler characteristic** is

$$\chi(K) = \sum_{k=0}^{d} (-1)^k f_k = f_0 - f_1 + f_2 - \cdots .$$

Example 1.9. The full simplex Δ^k has $f_j = \binom{k+1}{j+1}$, so

$$\chi(\Delta^k) = \sum_{j=0}^{k} (-1)^j \binom{k+1}{j+1} = 1 - (1-1)^{k+1} = 1,$$

as expected for a contractible space. Removing the top simplex gives the boundary $\partial\Delta^k \cong S^{k-1}$ with $\chi = 1 - (-1)^k = 1 + (-1)^{k-1}$, which is 2 for even-dimensional spheres and 0 for odd-dimensional ones.

1.2 Complexes from Data: The Nerve Theorem

In topological data analysis, we are often given a set of points (a point cloud) sampled from some unknown space, and we wish to construct a simplicial complex that approximates the topology of that space. The foundational theoretical tool for this is the Nerve Theorem.

Definition 1.10. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a finite collection of sets. The **nerve** of $\mathcal{U}$, denoted $N(\mathcal{U})$, is the abstract simplicial complex whose vertex set is I, and where a subset $\sigma \subseteq I$ is a simplex if and only if:

$$\bigcap_{i \in \sigma} U_i \neq \emptyset$$

The nerve is indeed a simplicial complex: if $\tau \subseteq \sigma$, then $\bigcap_{i\in\tau} U_i \supseteq \bigcap_{i\in\sigma} U_i \neq \emptyset$.

Theorem 1.11 (Nerve Theorem)**.** *If $\mathcal{U}$ is a finite collection of closed, convex sets in Euclidean space, then the nerve $N(\mathcal{U})$ is homotopy equivalent to the union of the sets $\bigcup_{i\in I} U_i$. More generally, the same conclusion holds for a finite open cover of a paracompact space, or a finite cover by subcomplexes of a simplicial complex, in which every non-empty intersection is contractible.*

Homotopy equivalence is weaker than homeomorphism but preserves everything we will compute in this course: the number of connected components, and later all homology groups.

Example 1.12. Cover the circle S^1 by three closed arcs U_1, U_2, U_3, each slightly longer than one third of the circle, so that consecutive arcs overlap but no point lies in all three. The pairwise intersections are small arcs (contractible) and the triple intersection is empty. The nerve is the hollow triangle on $\{1,2,3\}$, which is homeomorphic to S^1, in agreement with the Nerve Theorem.

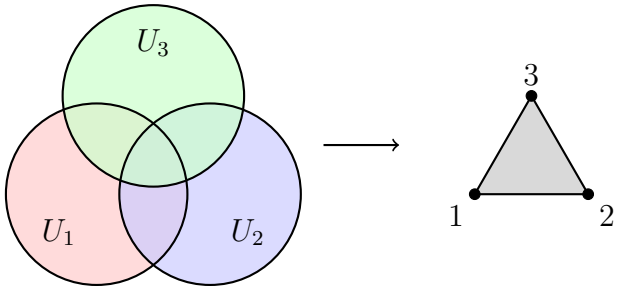


Figure 2: A cover $\mathcal{U} = \{U_1, U_2, U_3\}$ with a non-empty triple intersection (left) and its corresponding nerve $N(\mathcal{U})$ (right), which forms a solid 2-simplex.

Example 1.13. The contractibility hypothesis cannot be dropped. Cover S^1 by two closed arcs U_1 and U_2, each slightly longer than a half-circle. Each arc is contractible, but $U_1 \cap U_2$ consists of two disjoint small arcs, which is not contractible. The nerve is a single edge $\{1, 2\}$, which is contractible, while $U_1 \cup U_2 = S^1$ is not. The nerve has lost the hole.

1.3 The Čech Complex

Using the Nerve Theorem, we can construct a complex from a point cloud. Let $X = \{x_1, \ldots, x_N\}$ be a finite set of points in a metric space, and let $\epsilon > 0$ be a scale parameter.

Definition 1.14. The **Čech complex**, denoted $\check{C}_\epsilon(X)$, is the nerve of the collection of closed balls of radius ϵ centered at the points of X. That is, a set of vertices σ forms a simplex in $\check{C}_\epsilon(X)$ if:

$$\bigcap_{x_i \in \sigma} \overline{B}_\epsilon(x_i) \neq \emptyset$$

For $X \subset \mathbb{R}^n$, the closed balls are convex, so by the Nerve Theorem the Čech complex $\check{C}_\epsilon(X)$ is homotopy equivalent to the union of the balls $\bigcup \overline{B}_\epsilon(x_i)$. Concretely, σ is a simplex if and only if the points of σ fit inside a closed ball of radius ϵ, i.e., the radius of the smallest enclosing ball of σ is at most ϵ. However, checking for the simultaneous intersection of many balls in high-dimensional space is computationally expensive.

Remark 1.15. For two points x_i, x_j, the edge $\{x_i, x_j\}$ appears in $\check{C}_\epsilon(X)$ exactly when $d(x_i, x_j) \leq 2\epsilon$. For three points in $\mathbb{R}^n$ forming a triangle, the smallest enclosing ball is the circumscribed ball if the triangle is acute (or right), and the ball having the longest side as diameter if the triangle is obtuse. So the 2-simplex appears at $\epsilon = R$ (the circumradius) for acute triangles and at $\epsilon = \frac{1}{2}$(longest side) for obtuse ones.

1.4 The Vietoris-Rips Complex

To alleviate the computational burden of the Čech complex, we introduce a simpler complex that only relies on pairwise distances.

Definition 1.16. The **Vietoris-Rips complex**, denoted $VR_\epsilon(X)$, is the abstract simplicial complex whose simplices are all subsets $\sigma \subseteq X$ such that the diameter of σ is at most 2ϵ. Equivalently, σ is a simplex if for every pair of vertices $x_i, x_j \in \sigma$, the distance $d(x_i, x_j) \leq 2\epsilon$.

Remark 1.17. Some texts define the parameter for the Vietoris-Rips complex as the maximum pairwise distance δ. In that case, we would write $d(x_i, x_j) \leq \delta$. We use 2ϵ here to compare it directly with the Čech complex of radius ϵ.

The Vietoris-Rips complex is much easier to compute because it is a **clique complex** (or flag complex). Once the 1-skeleton (the graph of vertices and edges) is determined by pairwise distances, all higher-dimensional simplices are automatically determined by finding cliques (fully connected subgraphs) in the 1-skeleton.

Example 1.18. Let $X = \{0, 1, 3, 4\} \subset \mathbb{R}$ with the usual distance, and take $\epsilon = 1$, so that two points are joined when their distance is at most 2. The pairs within distance 2 are $\{0, 1\}$, $\{1, 3\}$ and $\{3, 4\}$; the pairs $\{0, 3\}, \{0, 4\}, \{1, 4\}$ are too far apart. The graph is a path $0-1-3-4$ with no triangles, so $VR_1(X)$ is this path, a contractible 1-dimensional complex. At $\epsilon = 1.5$ the pairs $\{0, 3\}$ and $\{1, 4\}$ (distance 3) are added, creating the cliques $\{0, 1, 3\}$ and $\{1, 3, 4\}$, so $VR_{1.5}(X)$ consists of two triangles sharing the edge $\{1, 3\}$. At $\epsilon = 2$ all pairs are joined and $VR_2(X)$ is the full 3-simplex.

1.5 Comparing Čech and Vietoris-Rips

While $VR_\epsilon(X)$ is computationally cheaper, it is not necessarily homotopy equivalent to the union of balls. However, the two complexes are closely intertwined.

Lemma 1.19 (Čech and Vietoris-Rips Interleaving)**.** *For any finite point set $X \subset \mathbb{R}^n$ and any $\epsilon > 0$, we have the following inclusions:*

$$\check{C}_\epsilon(X) \subseteq VR_\epsilon(X) \subseteq \check{C}_{\sqrt{2}\epsilon}(X)$$

Proof. If $\sigma \in \check{C}_\epsilon(X)$, there is a point z with $d(x_i, z) \leq \epsilon$ for all $x_i \in \sigma$, and the triangle inequality gives $d(x_i, x_j) \leq 2\epsilon$ for all $x_i, x_j \in \sigma$; hence $\sigma \in VR_\epsilon(X)$. For the second inclusion, let $\sigma \in VR_\epsilon(X)$, so σ has diameter $D \leq 2\epsilon$. Jung's Theorem states that a set of diameter D in $\mathbb{R}^n$ is contained in a closed ball of radius $D\sqrt{\frac{n}{2(n+1)}}$. Therefore σ lies in a ball of radius

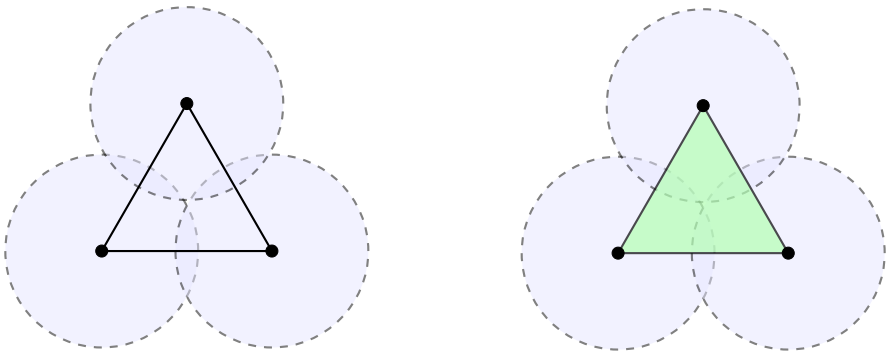
$$2\epsilon\sqrt{\frac{n}{2(n+1)}} = \epsilon\sqrt{\frac{2n}{n+1}} \leq \sqrt{2}\,\epsilon,$$

and its center lies in every $\overline{B}_{\sqrt{2}\epsilon}(x_i)$ with $x_i \in \sigma$. Hence $\sigma \in \check{C}_{\sqrt{2}\epsilon}(X)$. □

Note: The constant depends on the ambient space. In $\mathbb{R}^n$ the proof gives the sharper inclusion $VR_\epsilon(X) \subseteq \check{C}_{\epsilon\sqrt{2n/(n+1)}}(X)$ (for instance $\epsilon \cdot 2/\sqrt{3}$ in the plane), and $\sqrt{2}\epsilon$ is a bound valid in every dimension. In a general metric space, choosing the center to be any point of σ only gives $VR_\epsilon(X) \subseteq \check{C}_{2\epsilon}(X)$.

This interleaving property guarantees that persistent topological features found in the Vietoris-Rips complex closely approximate those of the Čech complex, making the VR complex the standard choice for computational topology applications.

Example 1.20. Let X be the four corners of the unit square, $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (1, 1)$, $x_4 = (0, 1)$. The side lengths are 1 and the diagonals have length $\sqrt{2}$. Any three of the points form a right triangle with hypotenuse $\sqrt{2}$, whose smallest enclosing ball has radius $\frac{\sqrt{2}}{2}$, and the same ball (centered at $(\frac{1}{2}, \frac{1}{2})$) encloses all four points. We obtain:



$\check{C}_\epsilon(X)$ (Hollow Triangle) $VR_\epsilon(X)$ (Solid Triangle)

Figure 3: For the given radius ϵ, pairwise intersections are non-empty, so both complexes contain all three edges. However, the triple intersection $\bigcap \overline{B}_\epsilon(x_i)$ is empty. Thus, the Čech complex is a hollow 1-cycle. The Vietoris-Rips complex, being a clique complex, automatically fills in the 2-simplex when all pairwise edges are present.

Scale	$\check{C}_\epsilon(X)$	$VR_\epsilon(X)$
$0 < \epsilon < \frac{1}{2}$	4 isolated points	4 isolated points
$\frac{1}{2} \leq \epsilon < \frac{\sqrt{2}}{2}$	hollow square (a 4-cycle) $\cong S^1$	hollow square $\cong S^1$
$\epsilon \geq \frac{\sqrt{2}}{2}$	full 3-simplex (contractible)	full 3-simplex (contractible)

The union of the four disks of radius ϵ has a hole in the middle precisely for $\frac{1}{2} \leq \epsilon < \frac{\sqrt{2}}{2}$, which the Čech complex detects exactly, as the Nerve Theorem predicts. Here the two complexes coincide at every scale. The loop is *born* at $\epsilon = \frac{1}{2}$ and *dies* at $\epsilon = \frac{\sqrt{2}}{2}$; recording such birth and death values across all scales is the idea behind persistent homology (Weeks 7–8).

Example 1.21. For the equilateral triangle of the figure above the two complexes differ. If the side length is s, the circumradius is $R = s/\sqrt{3}$, so

$$\check{C}_\epsilon \text{ is a hollow triangle for } \tfrac{s}{2} \leq \epsilon < \tfrac{s}{\sqrt{3}}, \qquad VR_\epsilon \text{ is a solid triangle for all } \epsilon \geq \tfrac{s}{2}.$$

The Čech complex sees a short-lived loop, born at $\frac{s}{2}$ and dying at $\frac{s}{\sqrt{3}} \approx 0.577s$, which the Vietoris-Rips complex misses entirely. Note that $\frac{s}{\sqrt{3}} = \frac{2}{\sqrt{3}} \cdot \frac{s}{2}$, so this example shows that the planar constant $\frac{2}{\sqrt{3}}$ in the sharper interleaving is attained.

Worksheet 1

1. **Continuity and the Discrete Topology:** Let X be a space equipped with the discrete topology (where every subset is open), and let Y be an arbitrary topological space. Prove that *any* function $f : X \to Y$ is continuous.

2. **Topological Invariants:** Prove that the closed interval $I = [0, 1]$ and the unit circle S^1 are not homeomorphic.

3. **Euler Characteristic and Connected Sums:**

   (a) Use the connected sum formula to compute the Euler characteristic of the surface $S = T^2 \# T^2 \# \mathbb{R}P^2$.

   (b) Is this resulting surface S orientable or non-orientable?

   (c) According to the Classification of Closed Surfaces theorem, which standard surface is S homeomorphic to?

4. **Abstract Simplicial Complexes:** Let $V = \{1, 2, 3, 4, 5\}$. Define an abstract simplicial complex K whose maximal simplices (facets) are $\{1, 2, 3\}$, $\{2, 3, 4\}$, and $\{4, 5\}$.

   (a) List *all* the simplices of K.

   (b) What is the dimension of the complex K?

   (c) Sketch the geometric realization $|K|$ in $\mathbb{R}^3$.

   (d) Compute the f-vector and the Euler characteristic of K.

5. **Čech vs. Vietoris-Rips Complexes:** Let $X = \{x_1, x_2, x_3\}$ be a point cloud in $\mathbb{R}^2$ consisting of three points that form an equilateral triangle with a side length of exactly $d = 2$.

   (a) Sketch the Vietoris-Rips complex $VR_\epsilon(X)$ for the scale parameter $\epsilon = 1$. Does it contain a 2-simplex?

   (b) Calculate the circumradius of this equilateral triangle.

   (c) Sketch the Čech complex $\check{C}_\epsilon(X)$ for the scale parameter $\epsilon = 1$. Does it contain a 2-simplex?

   (d) Verify that the interleaving lemma $\check{C}_1(X) \subseteq VR_1(X) \subseteq \check{C}_{\sqrt{2}}(X)$ holds for this specific point cloud.

6. **The Link of a Vertex:** Let K be an abstract simplicial complex. The *link* of a vertex v is defined as the subcomplex $\text{lk}(v) = \{\tau \in K \mid v \notin \tau, \tau \cup \{v\} \in K\}$. Prove that if the polyhedron $|K|$ is a connected closed surface, the geometric realization of the link of every vertex must be homeomorphic to the unit circle S^1.

7. **Minimal Triangulations:** Determine the minimal number of vertices V required to construct an abstract simplicial complex whose geometric realization is homeomorphic to the real projective plane $\mathbb{R}P^2$. Rigorously prove that no simplicial complex with fewer vertices can exist. Provide the exact number of edges E and faces F for this minimal triangulation.

8. **Polygonal Presentations:** Consider the hexagon with boundary word *abcacb*.

   (a) Determine the vertex classes after gluing and compute the Euler characteristic.

   (b) Decide whether the surface is orientable.

   (c) Identify the surface using the Classification Theorem.

9. **A Filtration of a Rectangle:** Let $X = \{(0,0),(2,0),(2,1),(0,1)\} \subset \mathbb{R}^2$. Describe $\check{C}_\epsilon(X)$ and $VR_\epsilon(X)$ for all $\epsilon > 0$, and determine the range of ϵ for which the complex is homotopy equivalent to a circle.